

# BIJOY SARKAR

Agartala, Tripura, India



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**Date of Birth**  
19-10-2005

**Nationality**  
INDIAN

**Gender**  
Male

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## PROFILE

Computer Science & Engineering undergraduate with hands-on experience in AI/ML, full-stack web development, cybersecurity, and mobile application development. Experienced in Python, Flutter, Firebase, computer vision, NLP, and software development through internships and practical projects. Currently seeking software development and AI/ML opportunities to build production-oriented solutions and strengthen problem-solving skills.

## EDUCATION

**Bachelor of Technology in Computer Science and Engineering**  
*ICFAI University Tripura, Agartala*

**Aug 2023 – Present**

## WORK EXPERIENCE

### Cybersecurity Intern

**Jun 2026 – Jul 2026**

*ETDC, STQC Directorate, MeitY, Government of India*

- Worked on cybersecurity frameworks and Generative AI security with focus on security assessment of AI applications.
- Conducted security risk analysis using **OWASP guidelines for Large Language Models (LLMs)** to identify potential vulnerabilities and mitigation strategies.
- Gained practical exposure to vulnerability assessment, AI application security, and IT testing and quality assurance practices.

### AI/ML Intern

**Jan 2026 – Mar 2026**

*National Institute of Technology Silchar*

- Worked on a **Road Segmentation** project using satellite imagery for identifying and extracting road networks.
- Performed image preprocessing, dataset preparation, feature extraction, and semantic segmentation using Python.
- Worked with satellite imagery datasets and evaluated computer vision models for road detection and segmentation.

### App Developer Intern

**Jun 2025 – Jul 2025**

*TalkEng, Agartala*

- Developed an **Expense Tracker** mobile application using **Flutter and Dart**.
- Implemented Firebase Authentication and real-time database integration for secure user data management.
- Worked on application functionality, user interface development, and overall user experience.

### Full-Stack Development Intern

**Jun 2024 – Aug 2024**

*Apni Estate, Agartala*

- Contributed to both front-end and back-end development of web applications.
- Worked on application design, coding, testing, debugging, and deployment-related activities.
- Collaborated on feature development and improved application functionality through practical development tasks.

### Computer Tutor

**Mar 2022 – Aug 2022**

*Novel Computer, Agartala*

- Taught computer fundamentals, programming basics, MS Office, and practical IT skills through hands-on learning.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, Java, JavaScript, Dart, SQL

**AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP, OpenCV, Jupyter Notebook

**Web & App:** HTML, CSS, Flask, Streamlit, Flutter, Firebase

**Databases:** MySQL, SQLite, Firestore

**Cybersecurity:** OWASP, LLM Security, Vulnerability Assessment, AI Security

**Tools:** Git, GitHub, Android Studio, VS Code

PROJECTS

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**OWASP AI Security Assistant** [GitHub]  
*Python Flask / Ollama / Llama 3 / Docker / OWASP*

- Developed a cybersecurity web application to assess code, URL endpoints, and infrastructure against OWASP vulnerability categories.
- Integrated local AI inference using **Ollama** and **Llama 3** for security analysis, vulnerability assessment, and remediation recommendations.
- Implemented assessment history, security metrics, dataset management, and re-evaluation workflows.
- Containerized the application using **Docker** and **Docker Compose** for portable and consistent deployment.

**Rakshak-AI**  
*AI Assistant / Python / LM Studio / Llama 3.2 / SQLite*

- Developed an AI-powered assistant using **Python** and a locally hosted LLM through **LM Studio**.
- Integrated conversational AI with a desktop GUI, BMI calculation, speech recognition, and text-to-speech functionality.
- Implemented **SQLite** for persistent data management and designed an interactive assistant experience.

**Road Segmentation from Satellite Images**  
*Python / Computer Vision / Jupyter Notebook / Kaggle*

- Developed a computer vision system for identifying and extracting road networks from satellite imagery.
- Performed image preprocessing, dataset preparation, model training, and evaluation using satellite image datasets.
- Explored semantic segmentation techniques for applications in geographic mapping and infrastructure planning.

**Expense Tracker**  
*Flutter / Dart / Firebase / Firestore*

- Developed a mobile application for recording and categorizing daily expenses using **Flutter** and **Dart**.
- Integrated **Firebase Authentication** and **Firestore** for secure user authentication and real-time data management.
- Implemented expense summaries and an intuitive user interface for monitoring personal spending.

**Smart Attendance System Using Face Recognition**  
*Python / OpenCV / Face Recognition / Computer Vision*

- Developed an automated attendance system using **face recognition** and **computer vision**.
- Implemented face detection, feature extraction, identity matching, and automated attendance recording.
- Designed the system to reduce manual effort and maintain organized attendance records.

ADDITIONAL PROJECTS

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**AI-Based Text Sentiment Analysis** *Python / Flask / NLTK / VADER / YouTube API*  
Developed a sentiment analysis application using Natural Language Processing techniques to classify textual content into positive, negative, and neutral sentiment categories.

**Advanced NLP** *Python / Jupyter Notebook / NLP / Kaggle*  
Developed an NLP project for processing and analyzing textual datasets, using preprocessing, tokenization, feature extraction, and text analysis techniques.

LANGUAGES

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English | Hindi | Bengali